

Describing Distributions 2

Description

- Location – mean and median
- Spread – Range of scores gives information about their spread
- Shape – symmetry, gaps, more/less dense regions, outliers, modality, convey information about the shape of a distribution
- Anything else of relevance

Any relevant information perhaps not covered in location, spread and shape?

Comment about location.

Comment about spread.

Comment on shape.

Anything else you notice of relevance.

Introduction

Open with a short description of the data.

Body

This is the section where you show how you interpret the distributions.

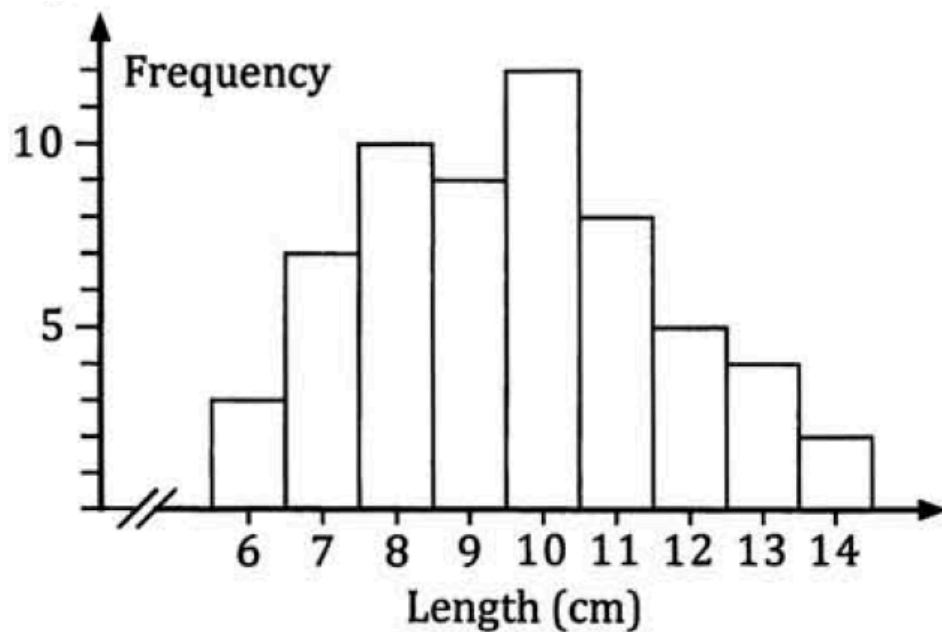
Conclusion (sort of)

State anything which you think is relevant.

Histogram vs Boxplot

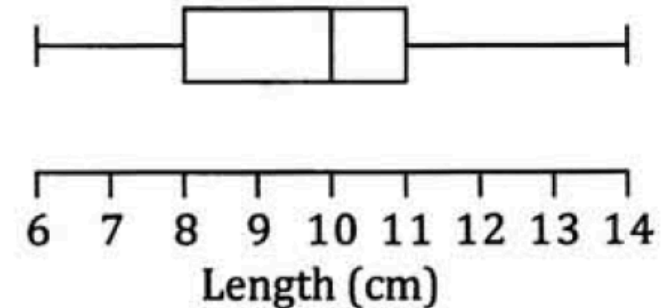
The lengths of the beaks of sixty male birds of a particular species were measured and the lengths, recorded to the nearest centimetre, were as follows:

8	7	10	13	11	9	9	7	10	12	8	11	7	11	8
11	13	9	14	10	7	6	10	8	11	10	8	9	10	7
9	12	11	8	6	8	10	12	9	7	10	8	12	6	9
7	10	14	9	8	10	13	8	10	12	11	11	13	9	10



Histogram:

- Convey the overall shape of the distribution
- Symmetry, grouping, gaps, modes etc.
- Can determine the mean and standard deviation.



Boxplot:

- Five number summary: median, lowest and highest scores, lower quartile and upper quartile.
- Range and interquartile range.
- Can draw several boxplots to be drawn in close proximity.

Description

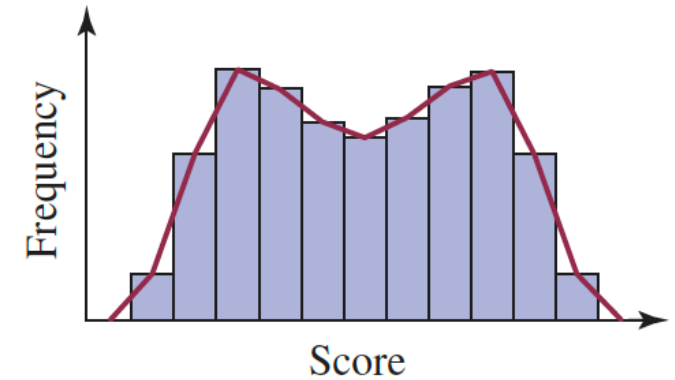
- Location – Where is it?
- Spread – How spread out is it?
- Shape – What does it look like?
- Anything else of relevance

Can now include Standard deviation.

- **Standard deviation** is a number used to tell how measurements for a group are spread out from the **mean**.
- A low standard deviation means that most of the numbers are very close to the **mean**.
- A high standard deviation means that the numbers are spread out.

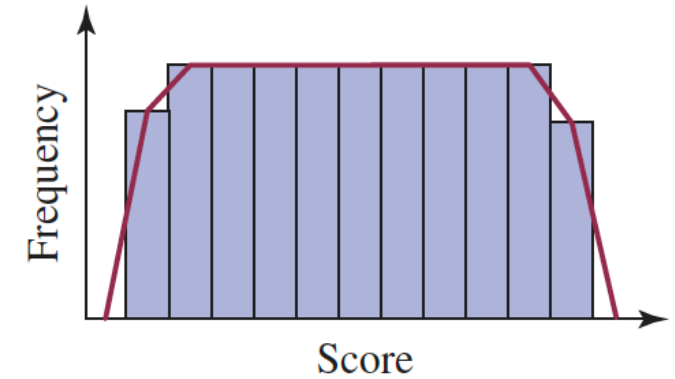
Bimodal data

There is more than one score that is most frequent.



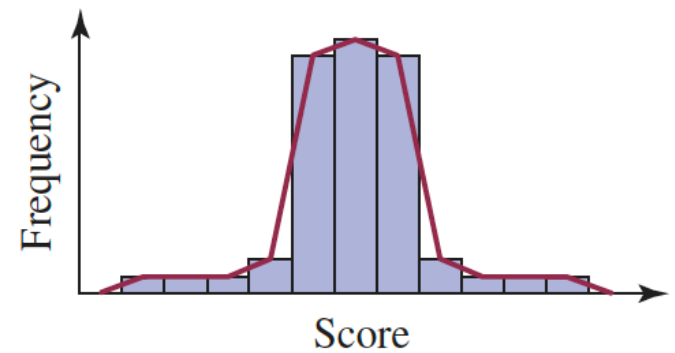
Spread data

The data are spread over a wide range.

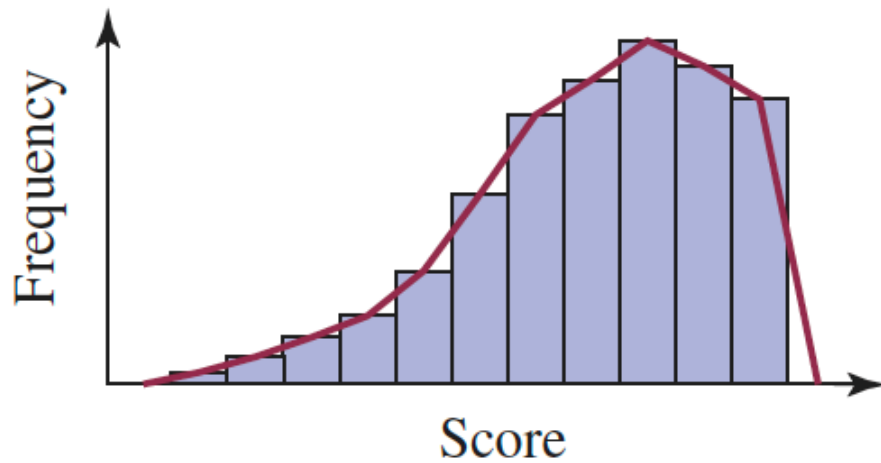


Clustered data

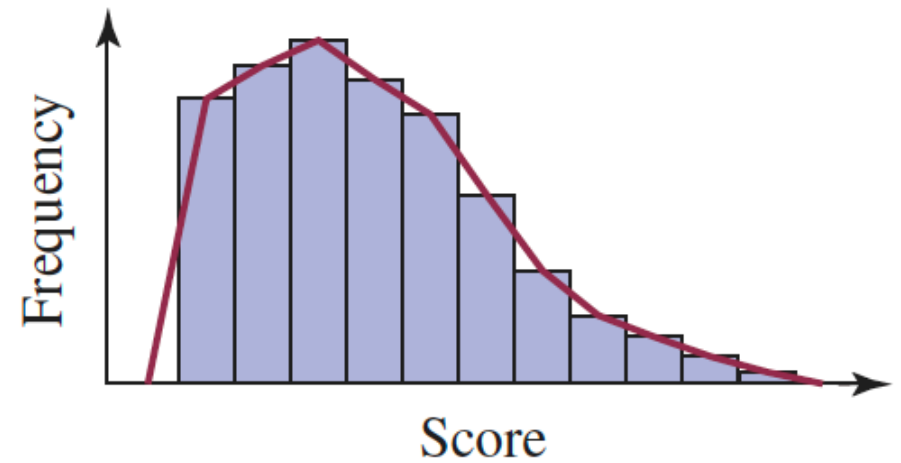
Most of the data are confined to a small range.



Skewness

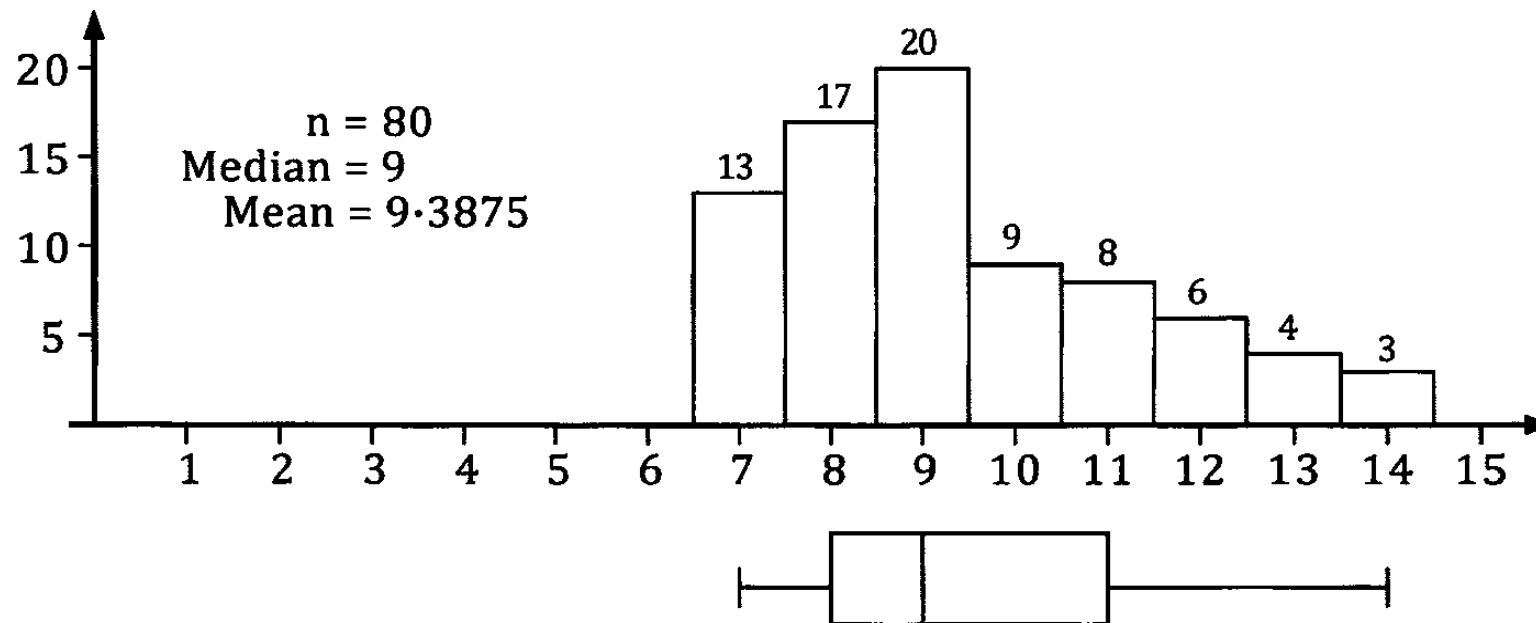


Negatively skewed
Skew to the left



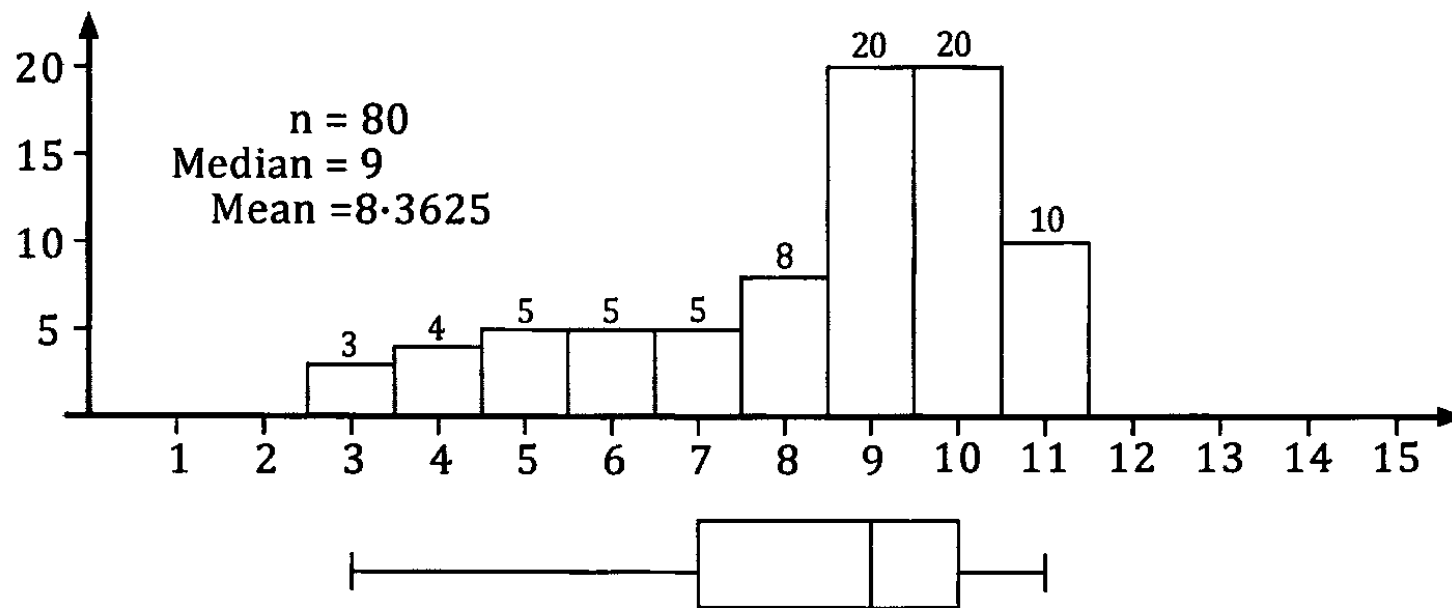
Positively skewed
Skew to the right

Positively skewed (skew to the right)

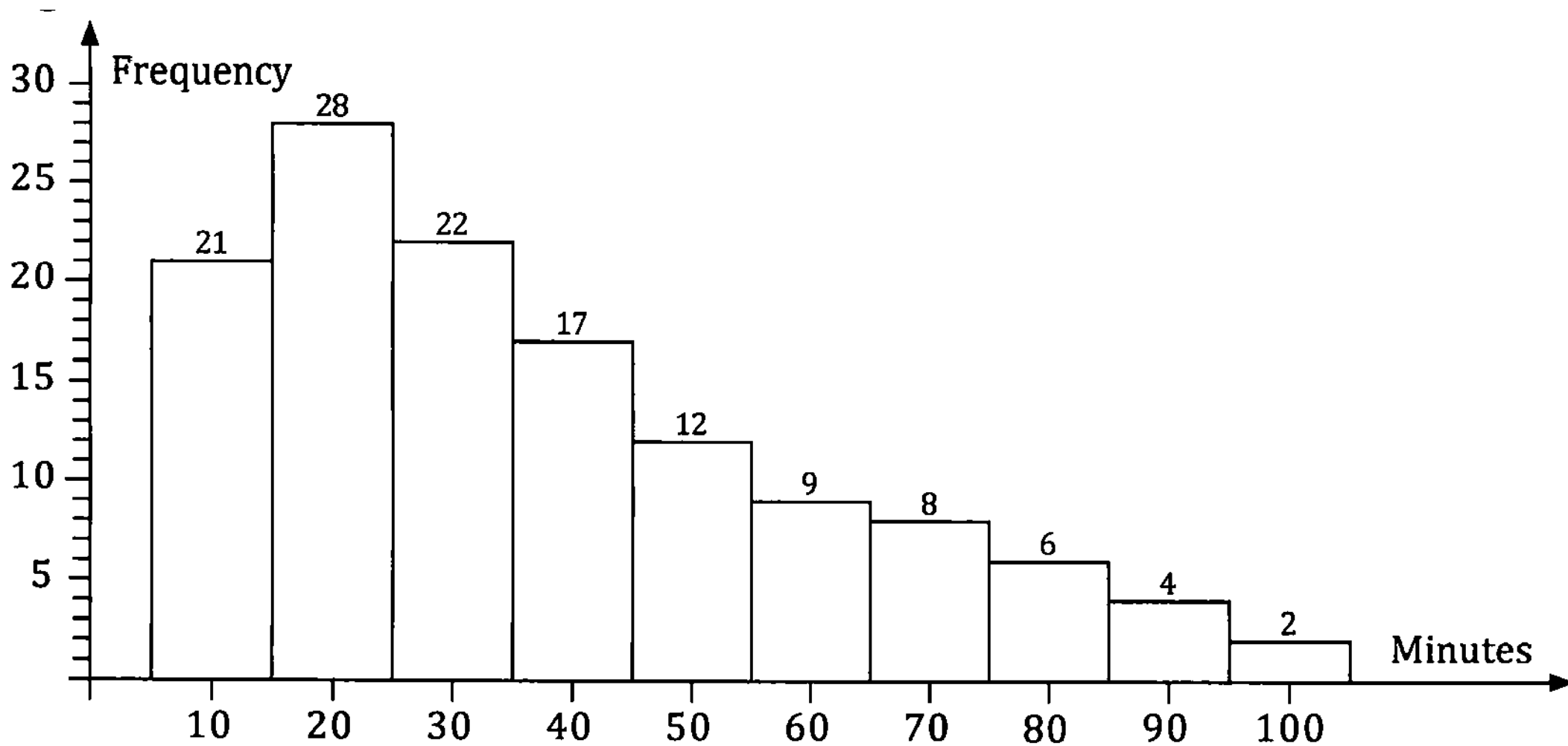


- Longer "tail" in the positive direction.
- Mean > median
- Boxplot will tend to be longer to the right of the median.

Negatively skewed (skew to the left)



- Longer "tail" in the negative direction.
- Median $>$ mean
- Boxplot will tend to be longer to the left of the median.



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